

Applying AmberMeta to Deposited MD Data

Standard Operating Procedure

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1 What this procedure is for

AmberMeta reads a directory of AMBER simulation files and reconstructs what was actually run: which topology belonged to which stage, which run continued which, how long each ran, and where the record is incomplete. The output is a *manifest* — a single YAML file that describes the campaign — plus optional summary artifacts.

This procedure covers depositing one *system* (one directory under the collection root). Work through it once per system. It uses the graphical interface throughout; every step has a command-line equivalent if you need to script it across many systems (§6).

The one thing to understand first

AmberMeta reports only what your files support. It will not invent a temperature it cannot read or assert a continuation it cannot demonstrate. If a deposit is missing its output logs, AmberMeta will say so rather than guess — which is the point, but it does mean an incomplete deposit produces a thin manifest.

2 Before you start: what a complete deposit contains

AmberMeta expects the file kinds below — the topology once per system, the rest once per run. The middle column is what is lost if that kind is absent.

File	What only this file can tell AmberMeta	Typical suffix
Topology	Atom count, box, whether masses were repartitioned	.prmtop, .top
Input	Requested settings: steps, timestep, thermostat, barostat	.mdin, .in
Output log	What <i>actually</i> ran: achieved time, temperature, completion, timings	.mdout, .out
Trajectory	Frame count, frame spacing, box volume over time	.nc, .mdcrd
Restart	The coordinates handed to the next run — the link in the chain	.restrt, .rst

The output log is the one most often missing and the most costly to lose. A deposit holding trajectories and restarts but no .mdout files can be assembled into a manifest, but nothing in it can be confirmed: every run's duration is unverified and every junction between runs is reported as an unexplained gap. Keep the logs.

Check before you deposit

Confirm every directory is readable (`chmod u+rx` on directories, not just files) and that no editor crash files, core dumps, or scratch copies are sitting among the run files. AmberMeta skips what it does not recognise, but a stray file that *looks* like a run — a lone .in in an analysis folder — becomes a step you then have to remove by hand.

3 Recommended layout for new work

For simulations you are about to run, this layout lets AmberMeta group everything correctly with no manual intervention:

```

system/
  topology/      system.prmtop, system.inpcrd
  minimization/  01_min.in, 01_min.out, 01_min.restrt
  equilibration/
    run1/        02_heat.in, 02_heat.out, . . .
    run2/
  production/
    run1/        prod_0001.in, prod_0001.out, prod_0001.nc, . . .
    run2/

```

Two rules carry all the weight:

1. **Replicas are sibling directories directly inside the stage they belong to**, named identically across stages (run1, run2, . . . or 01, 02, . . .). This is what lets AmberMeta recognise them automatically.
2. **The replica identifier appears in the directory name only, never in the filenames.** Use `prod/01/prod_0001.out`, not `prod/01/prod_01_0001.out`. Repeating the replica number inside the filename defeats automatic grouping (§5).

Chunked runs continue to use a numeric suffix (`prod_0001`, `prod_0002`); AmberMeta chains them by reading each run's restart, not by the numbering.

4 Procedure

4.1 Launch

Run AmberMeta at the top of the system directory you are depositing. That directory becomes a hard boundary — nothing outside it is read.

```
ambermeta gui /path/to/sys023
```

A browser opens on `http://127.0.0.1:8765`. The left pane lists the files it found, typed by what they are rather than by extension (Figure 1). Confirm the file counts look right before going further; if a whole stage is missing here, it is a permissions or path problem, not an AmberMeta one.

4.2 Discover

Press **Discover**. AmberMeta walks the tree, groups files into runs, infers each phase's role from the inputs, and links each run to the one whose restart it consumed. The result is a timeline (Figure 2).

Read down the timeline and check three things:

- **Phase roles.** Minimization, heating, equilibration and production should be labelled as such. A phase labelled *Unassigned* usually means a preparation step (tLEaP, ParmEd) was picked up as a run — set its role or delete it.
- **The chain.** Between consecutive steps you should see *restart of* A step reading *starting structure* part-way through a production sequence means the link could not be established.
- **Topologies.** Where several topologies exist, check each phase is bound to the right one. A repartitioned-mass topology is tagged HMR.

4.3 Define replicas

If the system was run as replicas, tell AmberMeta so. Runs in different replicas are independent trajectories; without this they are treated as one sequence and the junctions between them are reported as gaps.

Press **Define replicas** If the layout follows §3, the members are already proposed — accept them. Otherwise pick the directory level that distinguishes the replicas and accept only the members that really are replicas (preparation directories at the same level are offered too, and should be left alone).

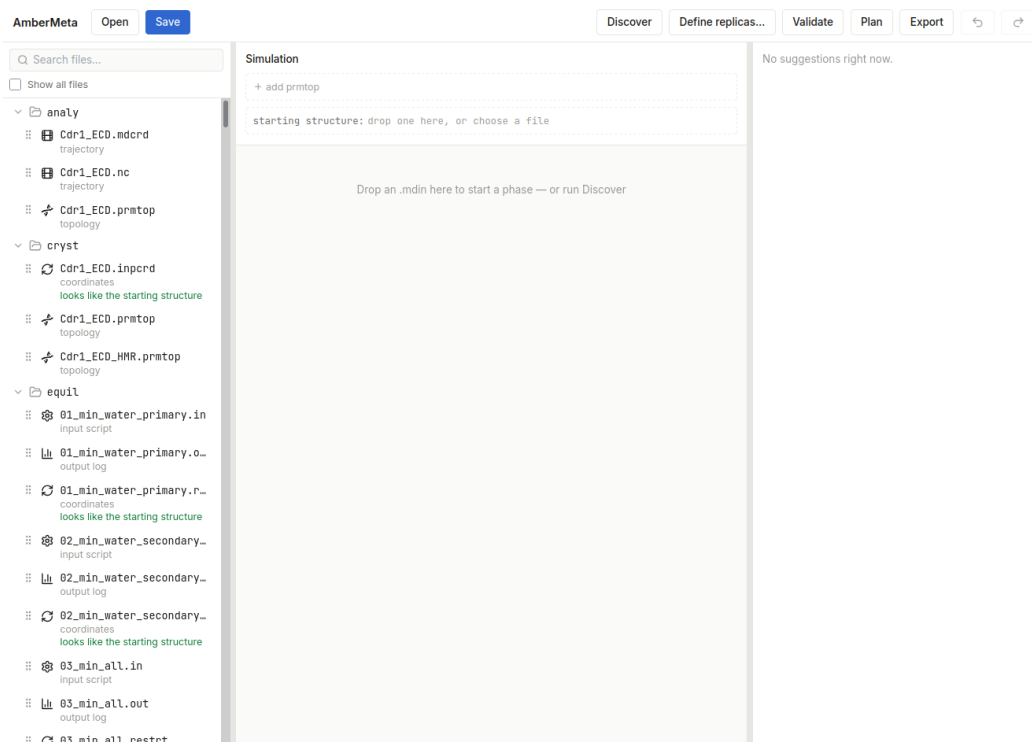


Figure 1: The file pane after launching against a system directory. Files are typed on sight — *topology*, *input script*, *output log*, *coordinates* — and candidates for the starting structure are flagged.

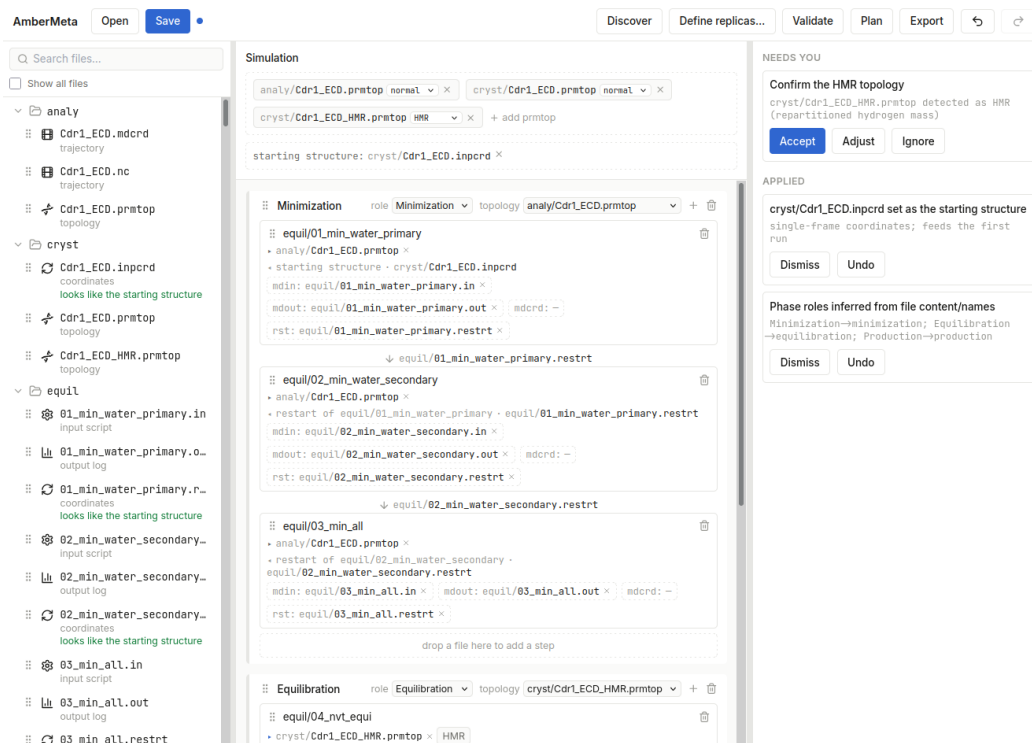


Figure 2: After **Discover**. The centre pane is the reconstructed timeline, with each run's inputs, outputs and the restart it continues from. The right pane separates what AmberMeta did on its own (**APPLIED**) from what it needs you to confirm (**NEEDS YOU**).

Once accepted, each phase carries a banner naming the member and its run count (Figure 3).

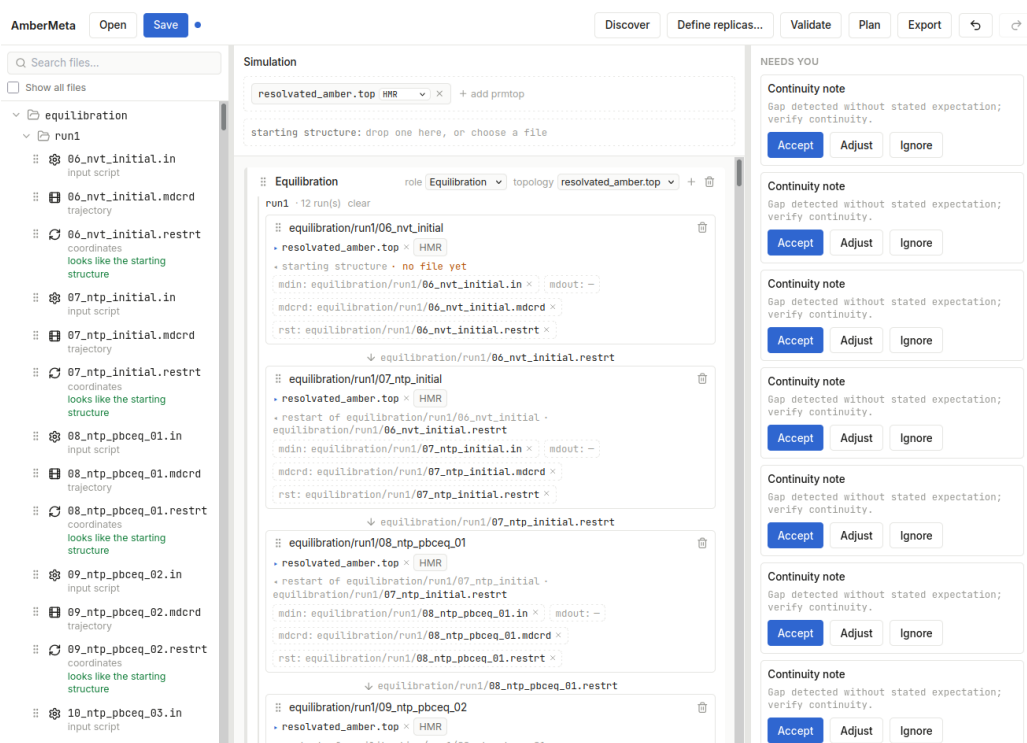


Figure 3: A system with replicas accepted: the run1 banner above the timeline marks the member and its twelve runs. The continuity notes on the right are this deposit's missing output logs — with no .mdout, no junction can be confirmed.

4.4 Validate, and resolve what it raises

Press **Validate**. Findings arrive in the right-hand tray in two groups:

APPLIED

Inferences AmberMeta already acted on, each with an **Undo**. Skim these — they are claims about your data. The starting structure and phase roles usually land here.

NEEDS YOU

Questions it will not answer on its own. **Accept** agrees, **Adjust** lets you supply the right value, **Ignore** records that you considered it.

The findings you are most likely to see, and what they mean:

Finding	What to do
Confirm the HMR topology	Check the flagged topology really is the mass-repartitioned one, then accept.
Gap detected without stated expectation	Two consecutive runs do not meet in time. Either a run is missing from the deposit, or the gap is real and deliberate — say which.
Atom count mismatch across files	A step is bound to the wrong topology. Rebind it in the timeline.
Trajectory is truncated or still being written	The run is incomplete or was copied mid-write. Re-copy it, or accept that its times come from the log instead.

Resolve findings until only ones you have deliberately accepted or ignored remain. A finding left unresolved is not an error — it is an unanswered question travelling with your data.

4.5 Save

Press **Save** to write `manifest.yaml` into the system directory. This is the deposit artifact. **Plan** additionally writes the summary files (`summary.json`, `methods_summary.json`) used for reporting.

Re-running this procedure later re-reads the files; unchanged files are not re-parsed, so a second pass over an unchanged system is quick.

5 Applying this to existing deposits

Data already collected does not follow one convention. The four encountered so far, and what each needs:

Layout	Example	Action
Replicas inside each stage	<code>equilibration/run1..run6</code>	Accept the proposal.
Numbered replicas in a stage	<code>prod/01..05</code>	Accept the proposal.
Replicas beside the stages	<code>run1..run4</code> beside <code>equilibration/</code>	Pick the top level in Define replicas... , then accept only <code>run1..run4</code> .
Replica id also in filenames	<code>prod/01/GB1_204_01_000</code>	Pick the replica level manually, then accept only the numbered members.

In the last two cases the automatic proposal is declined rather than wrong: AmberMeta refuses to guess when a layout admits more than one reading (Figure 4). Choosing the level yourself is the intended route, and the resulting manifest is identical to one from a conforming layout.

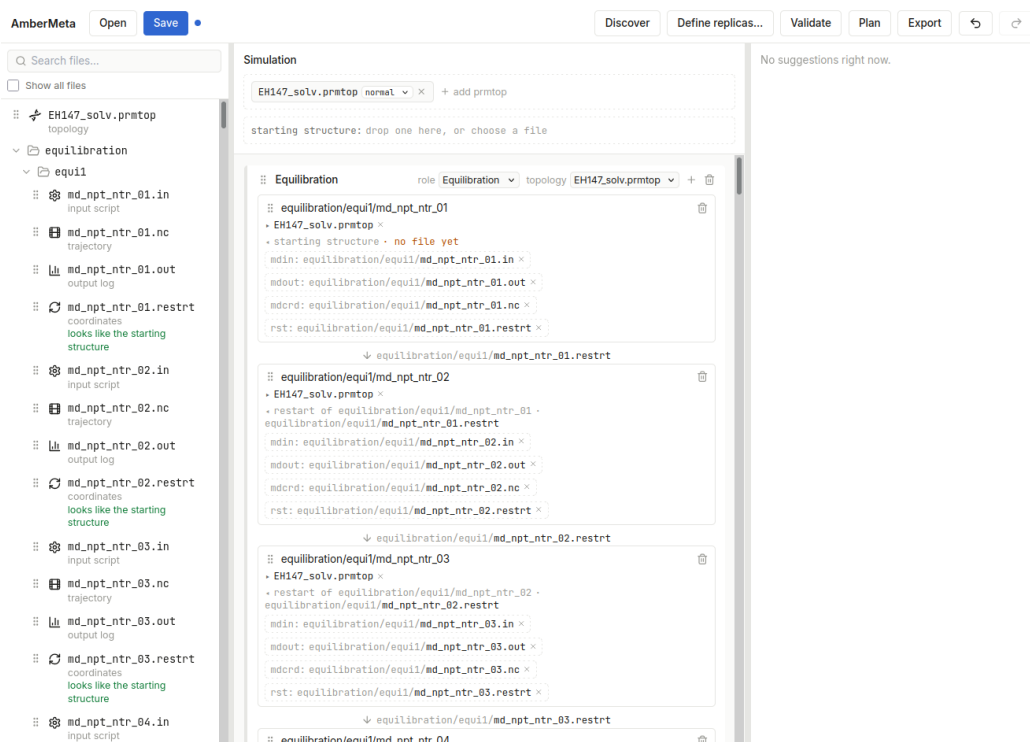


Figure 4: A layout where replicas were not proposed automatically: the runs are all present and correctly chained, but split across many phases with no member grouping. **Define replicas...** is what resolves this.

6 Quick reference

Step	Command-line equivalent
Launch	<code>ambermeta gui DIR</code>
Discover	<code>ambermeta discover DIR</code>
Discover + save	<code>ambermeta discover DIR --write manifest.yaml</code>
Build & check	<code>ambermeta plan DIR --recursive</code>
Summaries	<code>ambermeta plan DIR --recursive --summary-path summary.json</code>
Check one file	<code>ambermeta validate FILE [FILE ...]</code>
Inspect one file	<code>ambermeta info FILE</code>

Note that `validate` checks individual files; the whole-campaign check the **Validate** button performs is what `plan` does while assembling the protocol.

Checklist per system

1. Every directory readable; no stray files among the runs.
2. Output logs present for every run.
3. Discover run; phase roles and the restart chain read correctly.
4. Replicas defined, if the system has them.
5. Validate run; every finding accepted, adjusted or deliberately ignored.
6. `manifest.yaml` saved into the system directory.